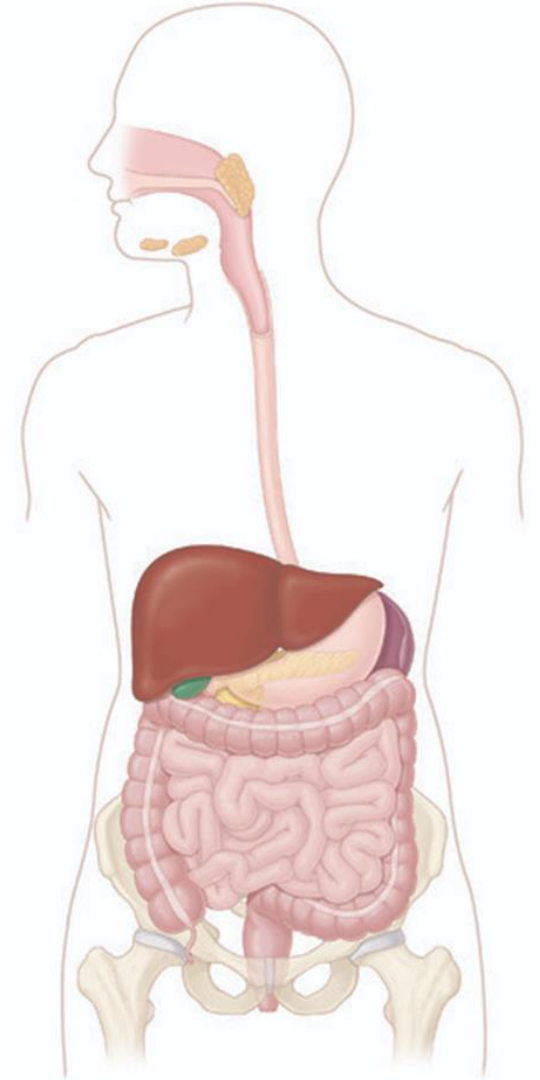




By:

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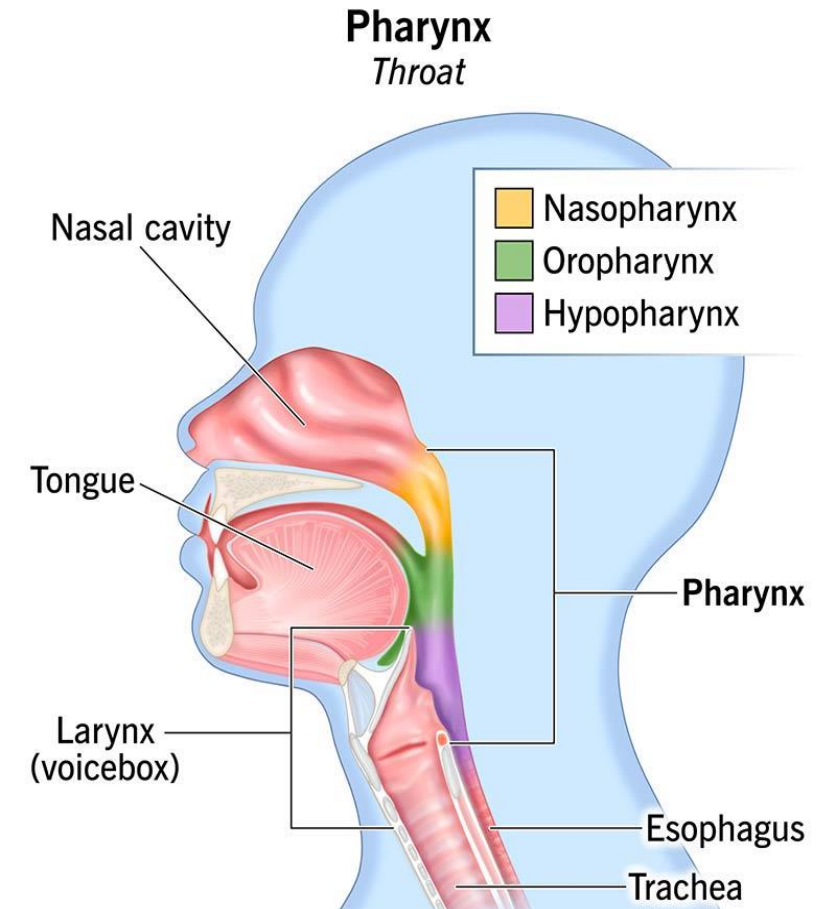
***Lec (2): Anatomy of the
Pharynx and Esophagus***

Objectives

- By the end of the lecture, the student should be able to:
 1. Describe the anatomical boundaries of the pharynx and differentiate its three regions.
 2. Identify and explain the key structures within each part of the pharynx, including the pharyngeal tonsil, palatine tonsil, and piriform recess.
 3. List and describe the parts of the esophagus, anatomical relations, and surface anatomy.
 4. Identify the esophageal constrictions and correlate them with relevant clinical conditions.
 5. The anatomy of lower esophageal sphincter, crural diaphragm & angle of His as protective factors that prevent the upward leak of the gastric contents

Pharynx

- ❑ The pharynx is a funnel-shaped fibromuscular tube, extending from the base of the skull to the esophagus (**lower border of C6**)
- ❑ The pharynx acts as a common channel for both food (deglutition) and air (respiration).
- ❑ *Length*: 12–14 cm, *Width*: 3.5 cm at its base, and 1.5 cm at pharyngoesophageal junction.
- ❑ Is divided into three parts. From above downwards these are as follows:
 - *Nasopharynx*, lying behind the nose.
 - *Oropharynx*, lying behind the oral cavity.
 - *Laryngopharynx*, lying behind the larynx.

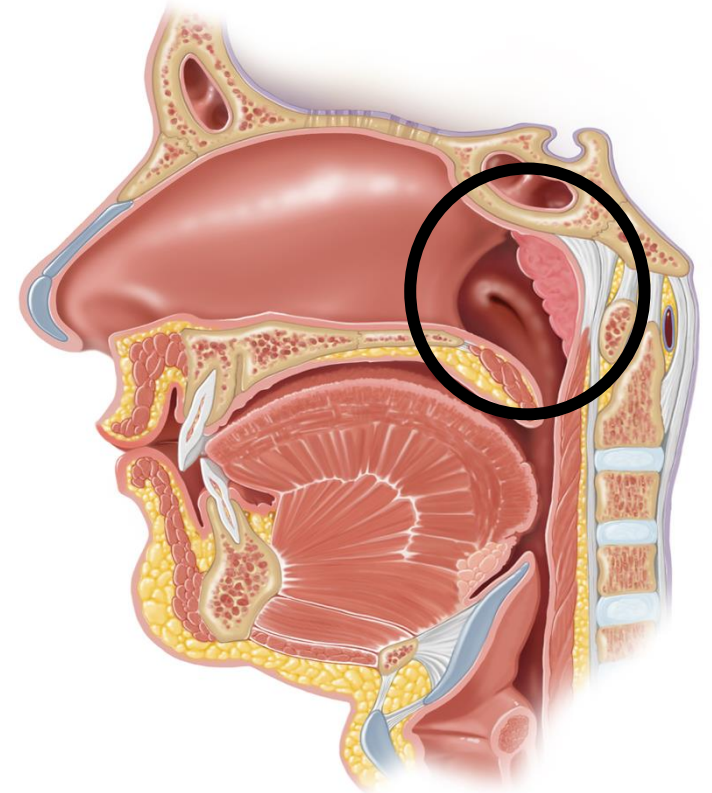
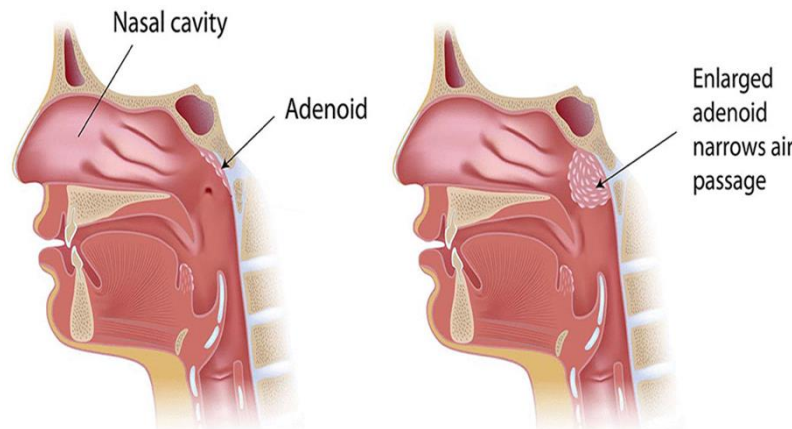


Nasopharynx

- ❑ The nasopharynx lies behind the nasal cavities and above the soft palate.
- ❑ Structurally and functionally the nasopharynx resembles the nose. It is respiratory in function and lined by pseudostratified ciliated columnar epithelium. Its walls are rigid and non-collapsible to keep the air passage patent

❑ Boundaries

- Anterior wall:.
- Posterior wall:
- Lateral wall
- Roof :



Oropharynx

The oropharynx provides common path for the food and air. It lies behind the oral cavity and extends from the lower surface of the soft palate above to the upper border of epiglottis below.

- **Boundaries**

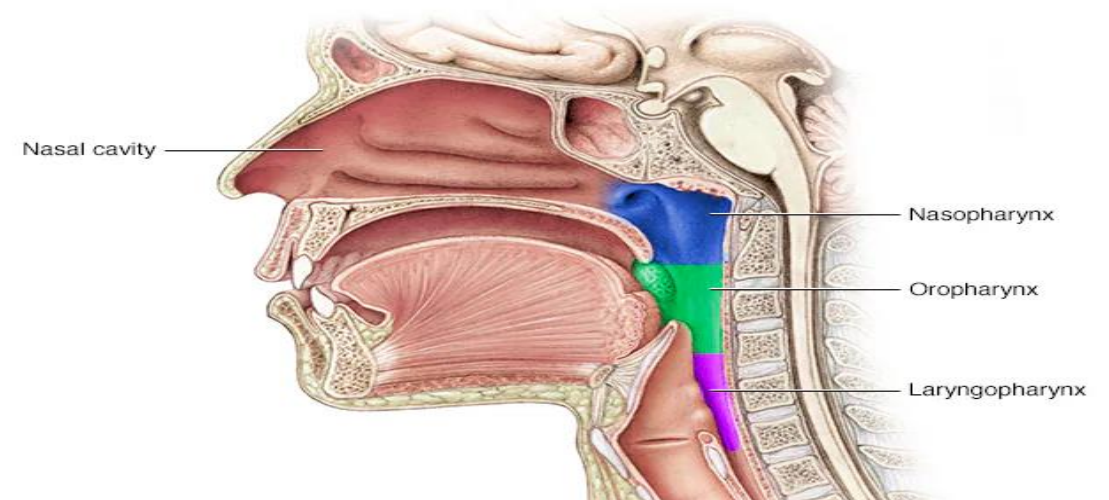
Roof: It is formed by: (a) Soft palate (under surface).

(b) Pharyngeal isthmus through which it communicated with the nasopharynx.

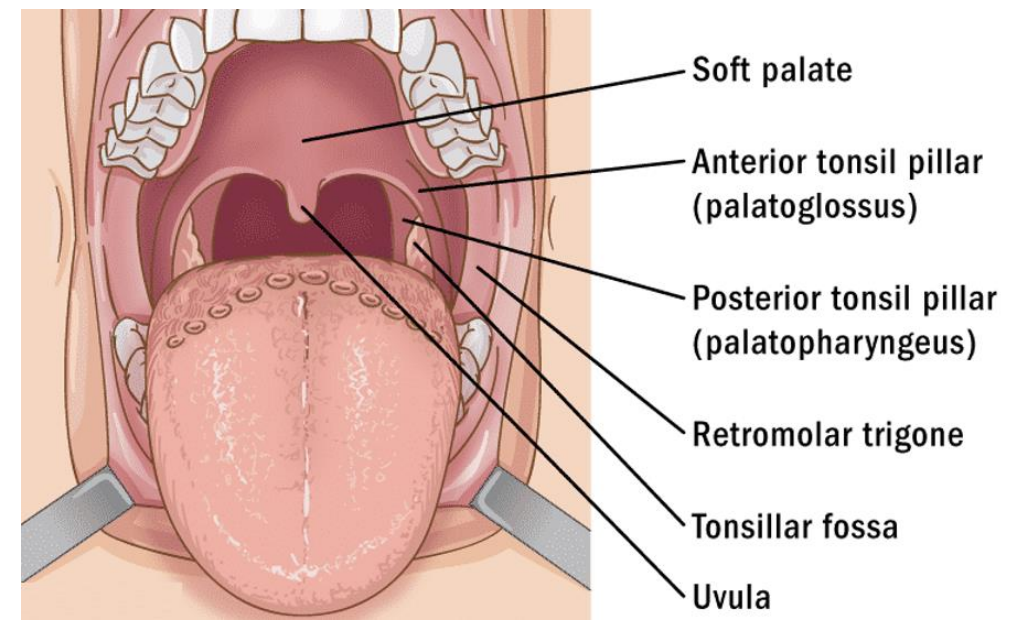
Floor: It is formed by: (a) Posterior 1/3rd of the tongue. (b) Interval between the tongue and epiglottis.

Anterior wall: It is incomplete and formed by: (a) Oropharyngeal isthmus (through which it opens into the oral cavity). (b) Pharyngeal part of the tongue.

Posterior wall: It is formed by body of C2 vertebra and upper part of the body of C3 vertebra.

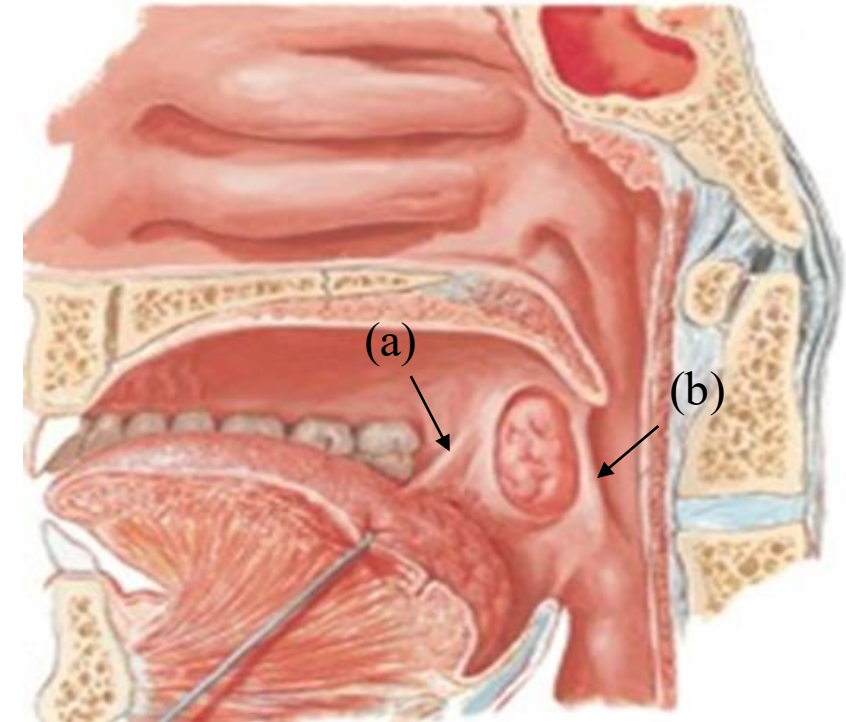


The oropharyngeal isthmus is closed during deglutition to prevent regurgitation of food from the pharynx to the mouth. Since the pathways for food and air cross each other in the oropharynx, the food sometimes may enter into the respiratory tract and cause choking. Similarly the air often enters the digestive tract producing gas in the stomach



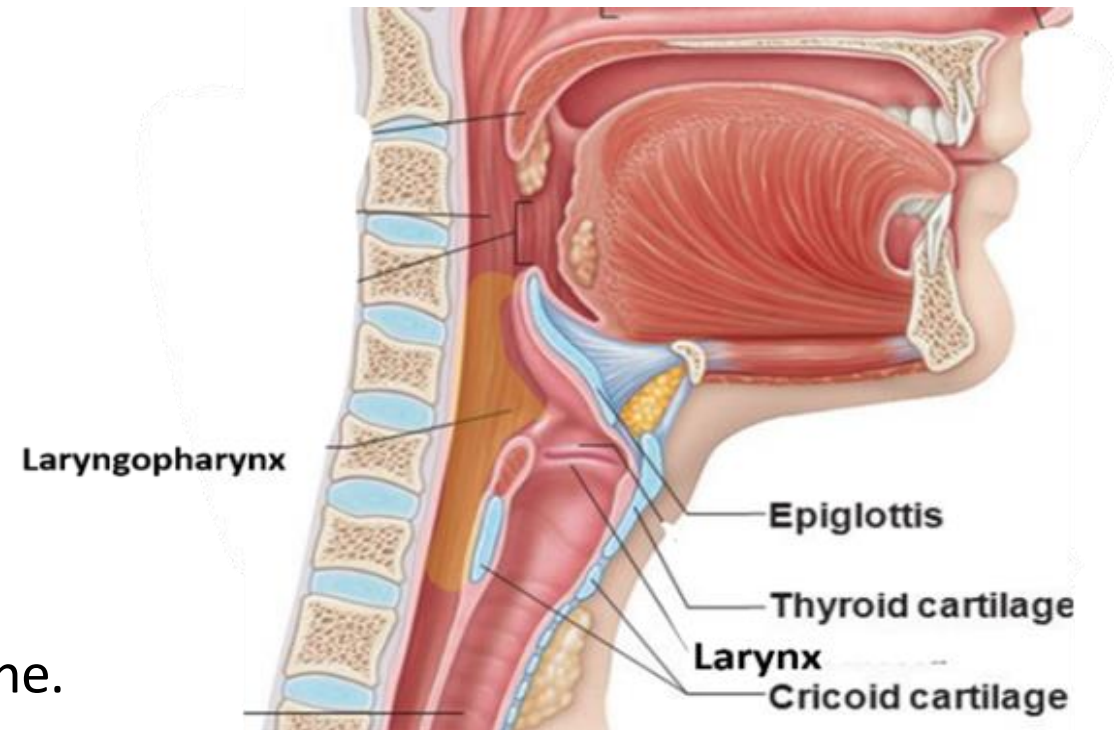
Lateral wall: On each side, it is supported by pterygomandibular raphe, mandible, tongue, and hyoid bone.

Lateral wall presents palatine tonsils, one on either side. It is located into a triangular fossa (tonsillar fossa) bounded anteriorly by (a) palatoglossal arch and (b) posteriorly by palatopharyngeal arch.



Laryngopharynx

- The laryngopharynx lies behind the laryngeal inlet and posterior wall of the larynx. It lies behind the larynx and extends from the upper border of the epiglottis to the lower border of cricoid cartilage anteriorly and lower border of C6 vertebra posteriorly. It communicates anteriorly with the laryngeal cavity through **laryngeal inlet** and inferiorly with the esophagus at the pharyngoesophageal junction (the narrowest part of the GIT except appendix).
- **Boundaries**
- *Anterior wall:* It is formed by: (a) Laryngeal inlet.
(b) Posterior surface of the larynx.
- *Posterior wall:* It is supported by the bodies of C3, C4, C5, and C6 vertebrae.
- *Lateral wall:* Thyroid cartilage and thyrohyoid membrane.



- **Piriform fossa**

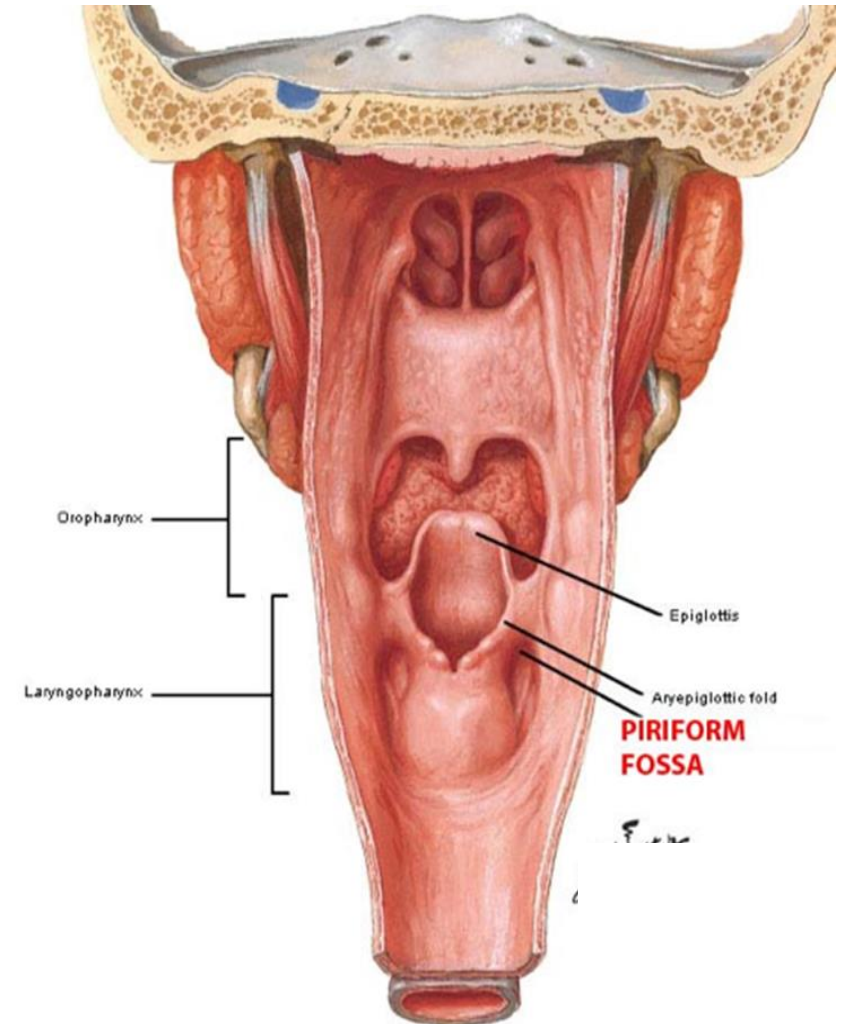
- It is a deep recess of lateral wall of the laryngopharynx, on each side of the laryngeal inlet. These recesses are produced due to bulging of larynx into laryngopharynx.

- **Boundaries**

- *Medial:* Aryepiglottic fold and quadrangular membrane of larynx.
- *Lateral:* Mucous membrane covering the medial surface of the lamina of thyroid cartilage and thyrohyoid membrane.

The internal laryngeal nerve and superior laryngeal vessels pierce the thyrohyoid membrane and traverse underneath the mucous membrane of the floor of the fossa to reach the medial wall.

- *Above:* Piriform fossa is separated from epiglottic vallecula by lateral glossoepiglottic fold.





Clinical Relevance

- ❑ The piriform fossae are dangerous sites for perforation by an endoscope.
- ❑ The ingested foreign bodies (for example, fish bones, safety pins) are sometimes lodged into the piriform fossa.
- ❑ If care is not taken, the removal of foreign bodies may damage the internal laryngeal nerve leading to anesthesia in the supraglottic part of the larynx and subsequent loss of *protective cough reflex*.



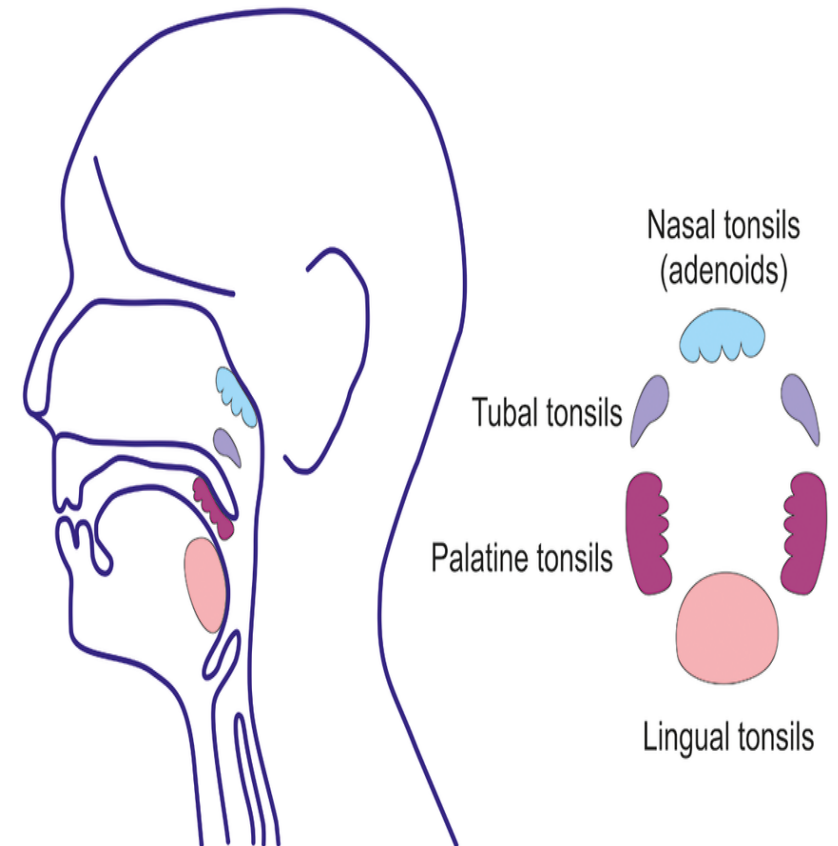
Waldeyer's ring

❑ The aggregations of lymphoid tissue underneath the epithelial lining of pharyngeal wall, surround the commencement of air and food passages. These aggregations together constitute an interrupted circle called *Waldeyer's ring*, which forms the special feature of the interior of the pharynx.

❑ The Waldeyer's ring is formed by

- **Pharyngeal tonsil (nasopharyngeal tonsil)**, postero-superiorly.
- **Lingual tonsil**, anteriorly.
- **Tubal and palatine tonsils**, laterally.

It prevents the invasion of microorganisms from entering the air and food passages and this helps in the defense mechanism of the Respiratory and alimentary systems.



Pharyngeal muscles

A. Constrictor Muscles

1. Superior constrictor

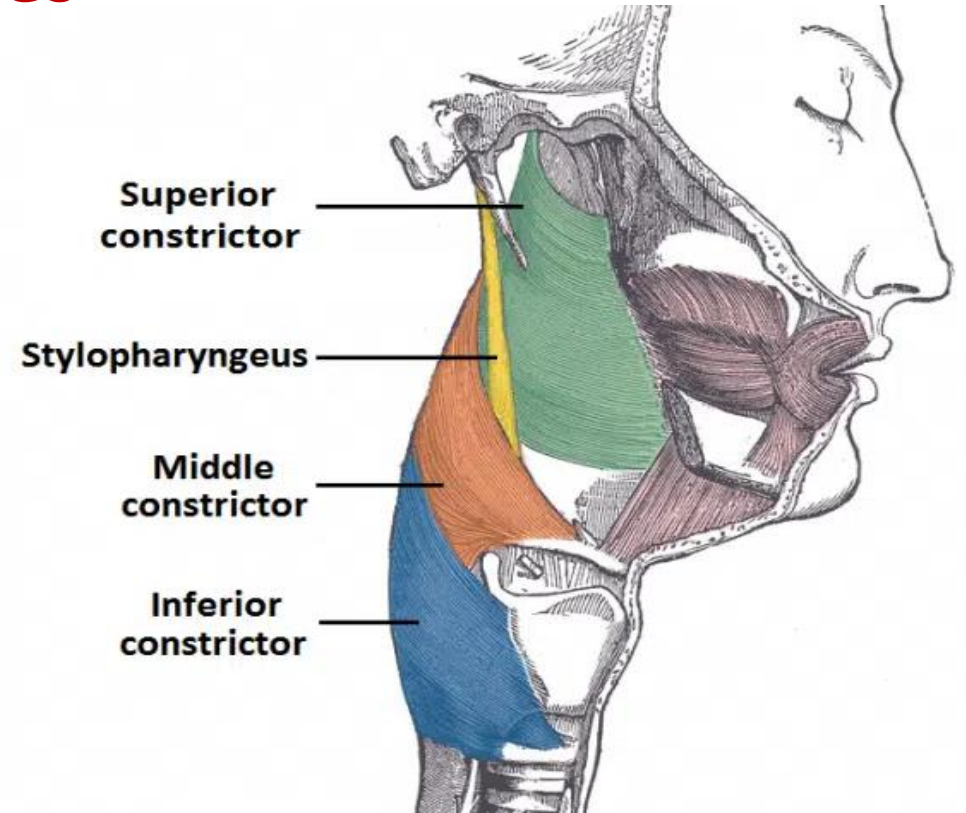
Origin: Pterygoid hamulus, Mylohyoid line of the mandible and Side of the tongue

Insertion: Pharyngeal raphe

Innervation: Pharyngeal plexus via **Vagus nerve (CN X)**

Action: Constricts the **upper part of the pharynx** during swallowing

Clinical Anatomy: Weakness may cause **nasopharyngeal regurgitation** during swallowing



2. Middle constrictor

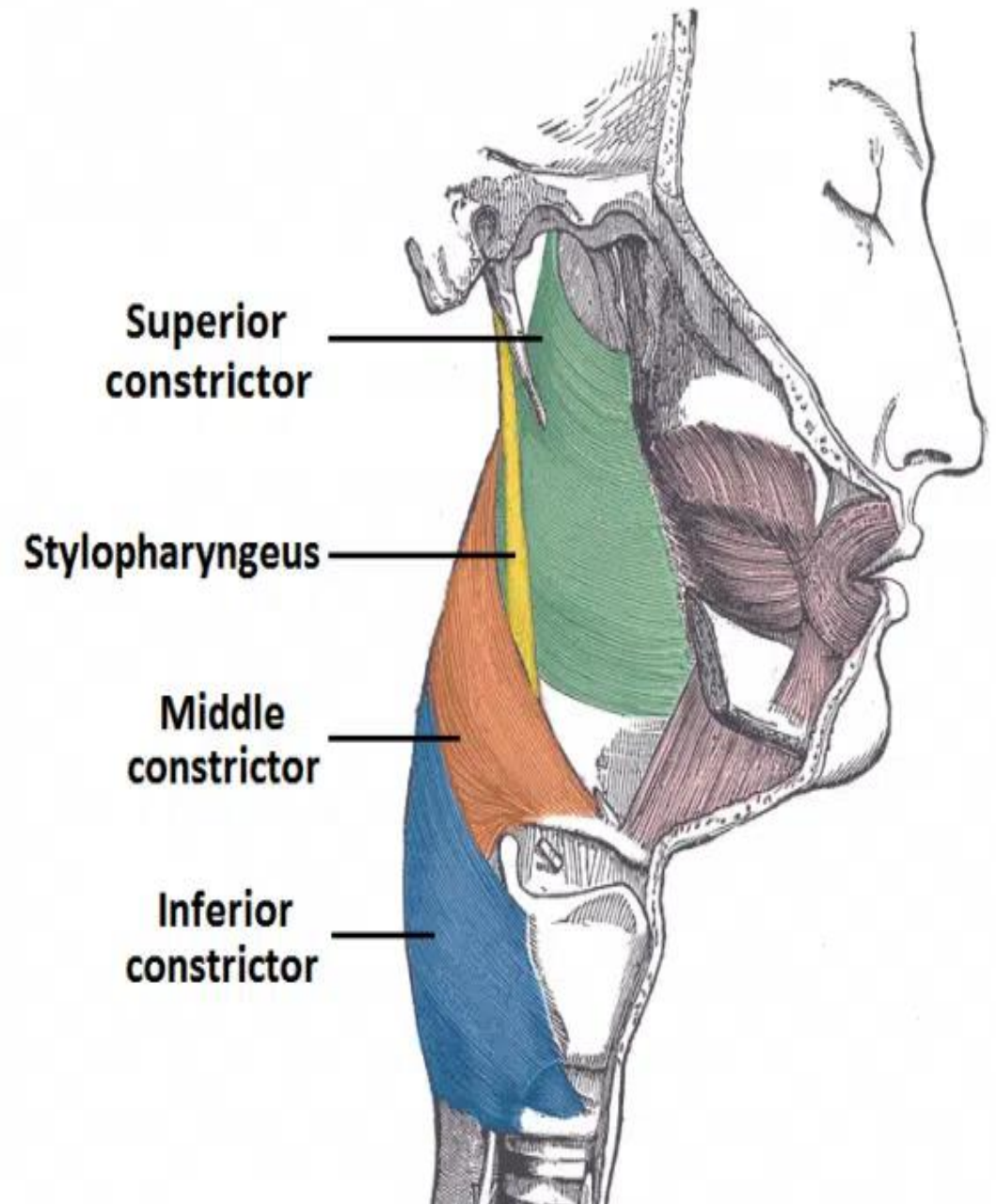
Origin: Greater and lesser horns of the hyoid bone

Stylohyoid ligament

Insertion: Pharyngeal raphe

Innervation: Pharyngeal plexus via Vagus nerve

Action: Constricts the middle part of the pharynx,
swallowing



3. Inferior constrictor

Origin:

Thyropharyngeus: oblique line of thyroid cartilage

Cricopharyngeus: cricoid cartilage

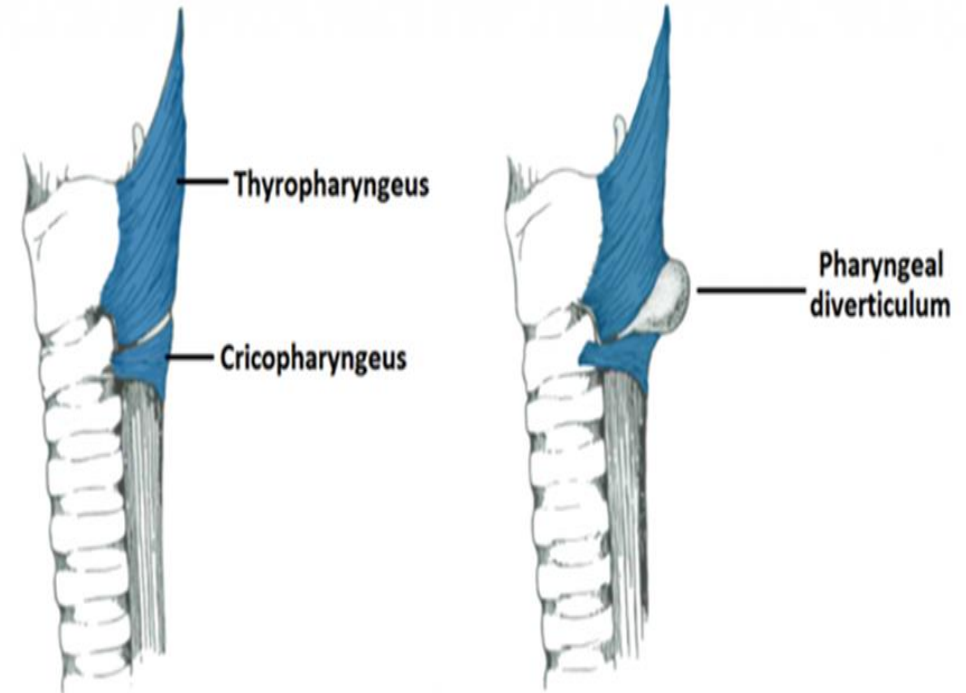
Between them (**Killian's dehiscence**) which is a **triangular weak area** in the posterior wall of the pharynx,

Insertion: Pharyngeal raphe

Innervation: Pharyngeal plexus via Vagus nerve

Action: Constricts lower pharynx Cricopharyngeus acts as the upper esophageal sphincter (UES), preventing air entry into the esophagus during breathing

Clinical Anatomy: common site for (a posterior herniation at Killian's dehiscence) with dysphagia





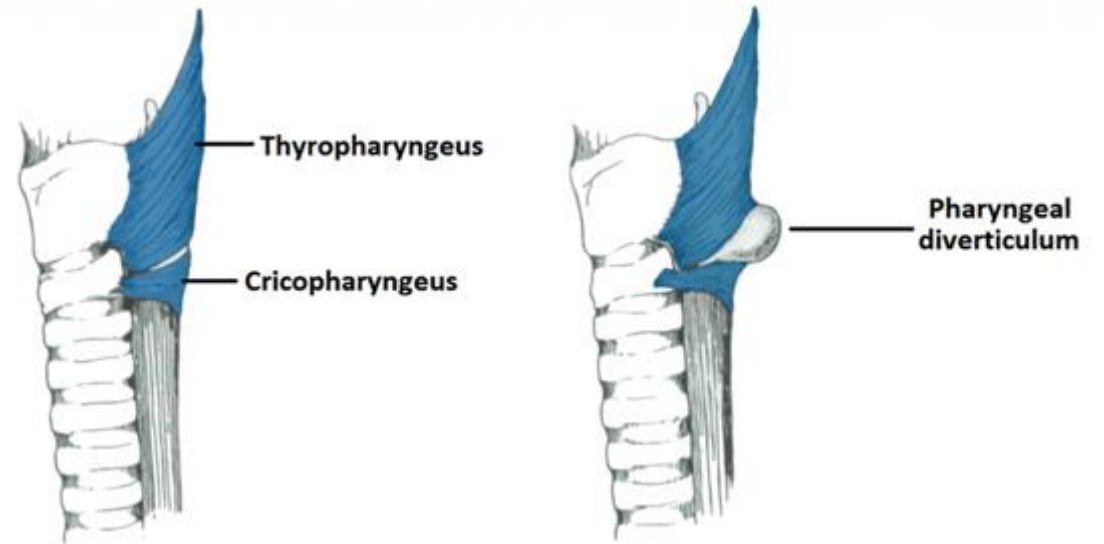
Clinical Relevance

Pharyngeal Diverticulum (Pouch)

The **inferior pharyngeal constrictor** is split into two parts; the

- Thyropharyngeus and the
- Cricopharyngeus.

This area between two muscles is a weak area in the mucosa. It is possible for food to accumulate here called diverticulum and leading to dysphagia





CLINICAL

CASE

SCENARIO



A 54-year-old man presents with severe throat pain after swallowing a fish bone. Imaging shows the foreign body lodged at the pharyngoesophageal junction, the point where the pharynx terminates. At which vertebral level is this foreign body impacted?

- A) Lower border of C3
- B) Lower border of C4
- C) Upper border of C5
- D) Lower border of C6
- E) Upper border of T2

A 5-year-old boy is brought to the clinic due to chronic mouth breathing, loud snoring, and persistent nasal obstruction. Examination confirms excessive hypertrophy of the lymphoid tissue located in the roof of the nasopharynx. Which tonsil is enlarged in this condition?

- A) Palatine tonsil
- B) Lingual tonsil
- C) Tubal tonsil
- D) Pharyngeal tonsil
- E) Laryngeal tonsil



CLINICAL

CASE

SCENARIO



A 7-year-old child presents with a sore throat, high fever, and painful swallowing. Oral examination reveals enlarged palatine tonsils. Between which two muscular folds are these tonsils located?

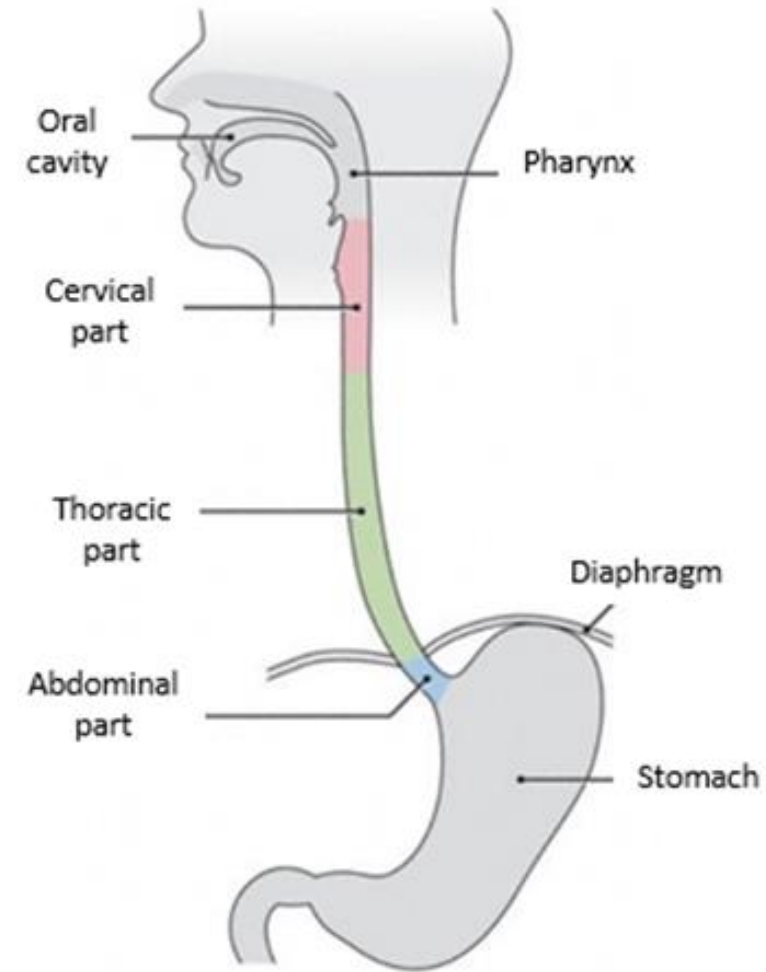
- A) Salpingopharyngeal and palatopharyngeal arches
- B) Palatoglossal and palatopharyngeal arches
- C) Palatoglossal and salpingopharyngeal arches
- D) Superior and middle pharyngeal constrictor muscles
- E) Stylopharyngeus and palatopharyngeus arches

A 19-year-old college student presents with severe unilateral throat pain. Examination found swelling of the palatine tonsil. In which division of the pharynx are the involved tonsils primarily located?

- A) Roof of the nasopharynx
- B) Lateral wall of the nasopharynx
- C) Lateral wall of the oropharynx
- D) Posterior wall of the laryngopharynx
- E) Piriform recess of the hypopharynx

ESOPHAGUS

- The esophagus is a muscular tube 10 inches (25 cm) long, extending from the laryngopharynx to the stomach.
 - It begins at the level of the cricoid cartilage (C6); it descends through the neck and continues through the superior mediastinum, then the posterior mediastinum,
 - It pierces the diaphragm at the level of T10.
 - It joins with the stomach at the cardiac orifice, at the level of T11.
- The esophagus divides into three parts (cervical, thoracic, and abdominal parts).



CERVICAL PART:

❑ This is the uppermost segment of the esophagus, measuring 4 cm, and is located in the neck

❑ It extends from the hypopharynx to the sternal notch.

❑ Relations:

○ **Posteriorly:** The 6th and 7th cervical vertebrae

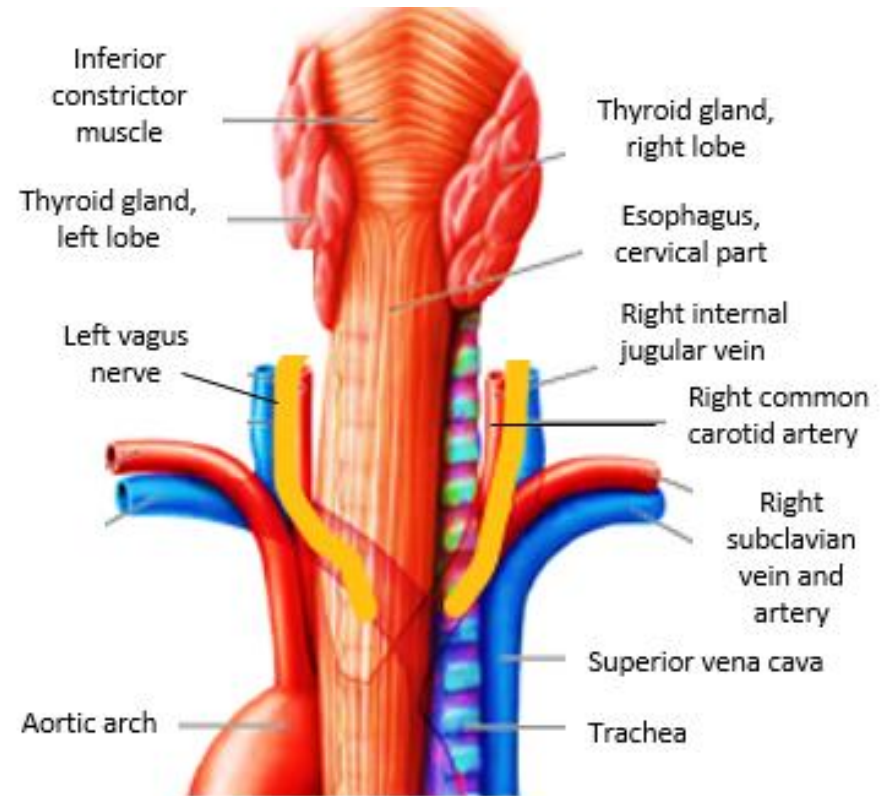
○ **Anteriorly:** The trachea and the recurrent laryngeal nerve.

○ **Laterally:** The thyroid lobes and carotid sheath

❑ **Blood Supply:** Supplied by the inferior thyroid artery and vein.

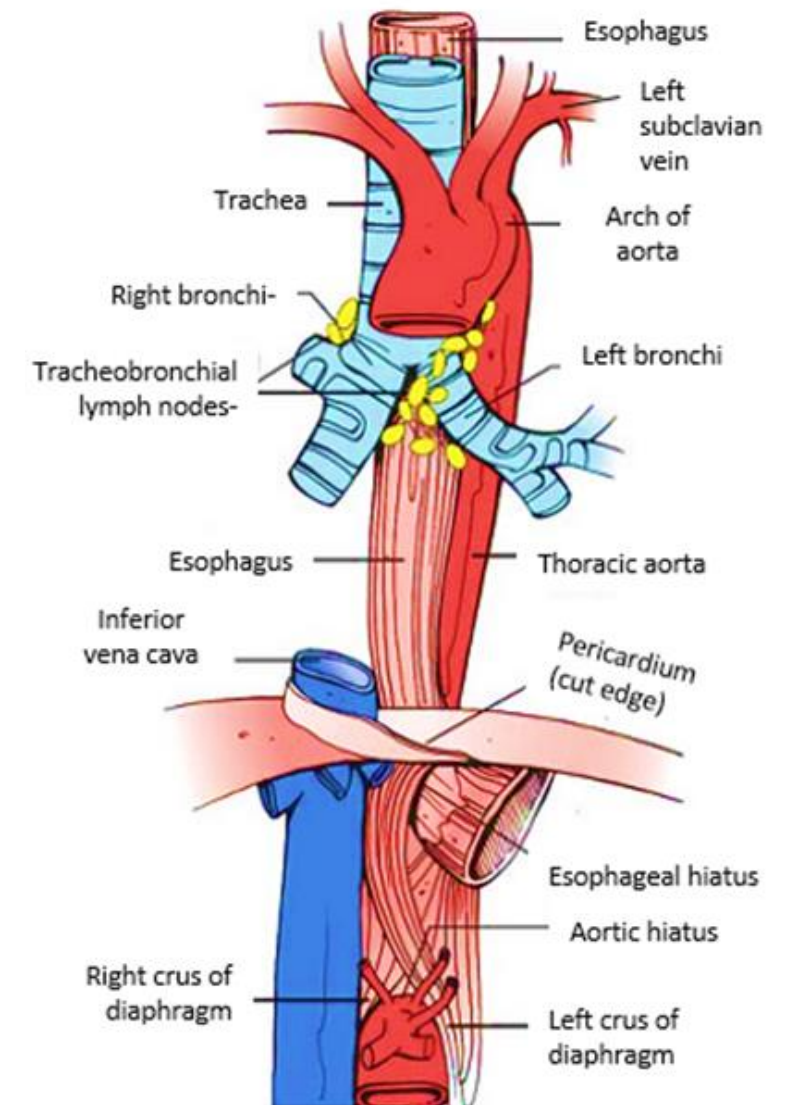
❑ **Nerve Supply:** Innervated by the recurrent laryngeal nerves

❑ **Lymph Drainage:** Drains into the deep cervical lymph nodes.

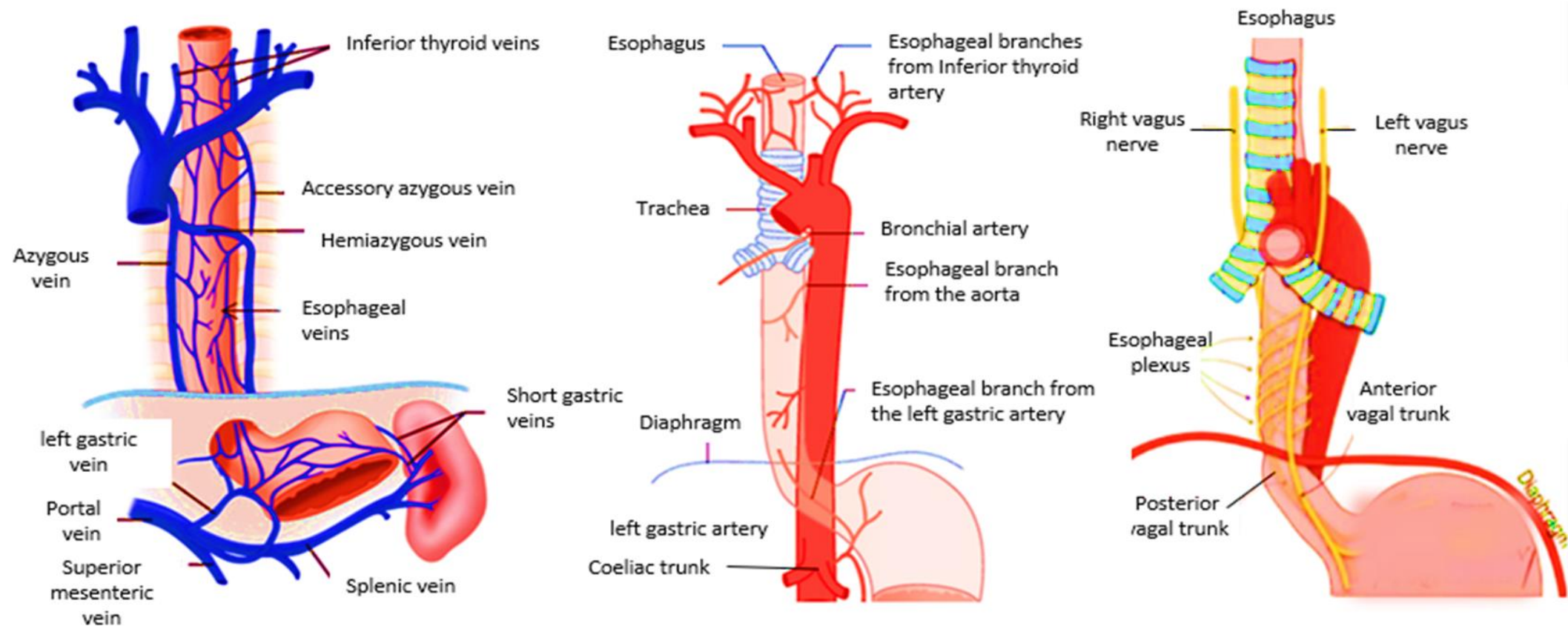


THORACIC PART

- ❑ Is the longest part of the esophagus 20 cm.
- ❑ It extends from the sternal notch (T1) to the level of T10.
- ❑ Related
 - **Anteriorly:** The trachea and the left recurrent laryngeal nerve;
 - **Posteriorly:** The bodies of the thoracic vertebrae, the thoracic duct, the azygos veins, and the descending thoracic aorta
 - **Right side:** The mediastinal pleura
 - **Left side:** The left subclavian artery, the aortic arch, the thoracic duct



- ❑ Blood Supply: esophageal branches from the descending thoracic aorta.
- ❑ Venous drainage: by the third of the veins of the veins into the azygos veins
- ❑ Lymph Drainage: into the superior and posterior mediastinal nodes.
- ❑ Nerve Supply: by the parasympathetic and sympathetic fibers via the vagi



A. Venous drainage **B.** Arterial supply and **C.** Nerve supply of three parts of the esophagus

ABDOMINAL PART:

- ❑ The abdominal part of the esophagus is the shortest segment, 1.5 to 3 cm
- ❑ It is located anteriorly to the left lobe of the liver and the left crus of the diaphragm.
- ❑ This part receives its blood supply from the left gastric artery, which is a branch of the celiac trunk, as well as from the left inferior phrenic artery
- ❑ Venous drainage for this portion of the esophagus occurs through two pathways:
 - To the portal circulation via the left gastric vein
 - To the systemic circulation via the azygous vein
 - These two pathways create a porto-systemic anastomosis, which is a connection between the portal and systemic venous systems.

- ❑ Esophageal varices refer to abnormally dilated submucosal veins (tributaries of the portal and azygos) within the wall of the esophagus, a condition often resulting from hypertension, which can lead to hematemesis.
- ❑ Lymphatic drainage occurs into the posterior mediastinal and left gastric lymph nodes.
- ❑ Nerve Supply: includes both sympathetic via the greater splanchnic nerves (T5-T9) and parasympathetic by the vagal trunks

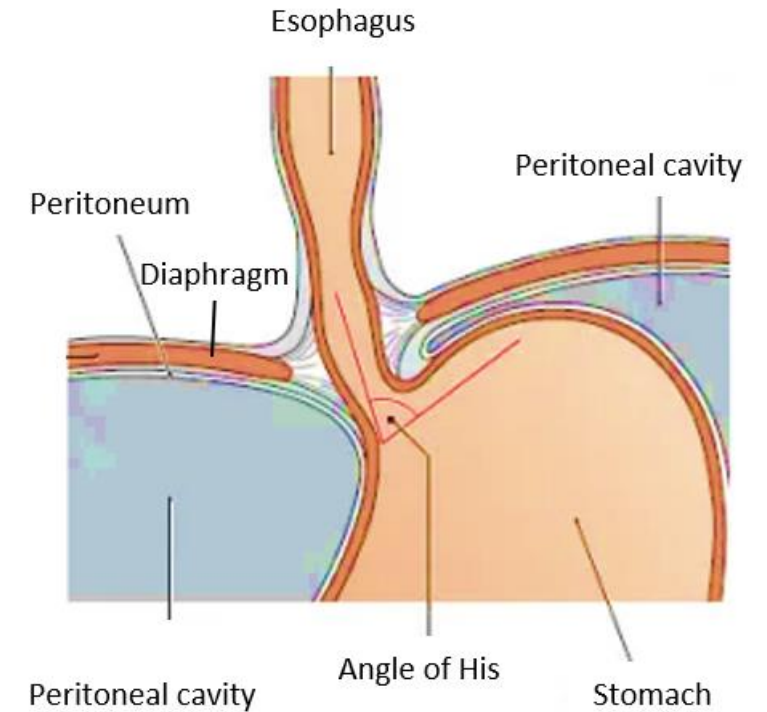
Esophageal Sphincters

❑ There are two sphincters in the esophagus they prevent the entry of air and the reflux of gastric contents

❑ **Upper Esophageal Sphincter:** is an anatomical, striated muscle sphincter located at the junction of the pharynx and esophagus, formed by the cricopharyngeus muscle

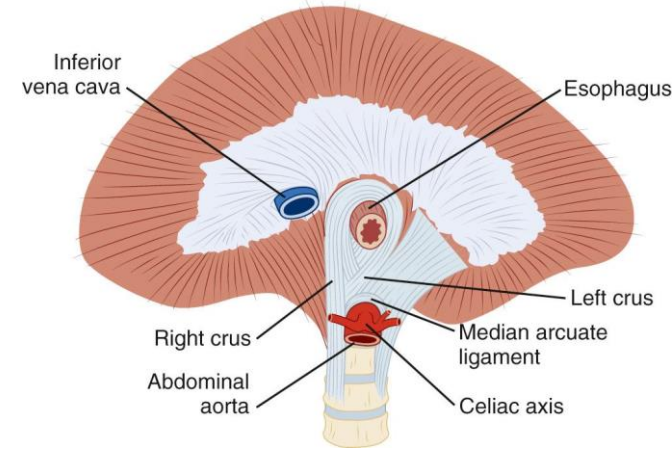
❑ **Lower Esophageal Sphincter:** is a physiological sphincter situated at the gastroesophageal junction (T11) provides the principal internal sphincteric resistance, is influenced by the right crus of the diaphragm and the gastroesophageal angle (angle of His)

❖ *When LES pressure is reduced or inappropriate transient LES relaxations occur, gastric contents can move upward into the esophagus, contributing to GERD.*

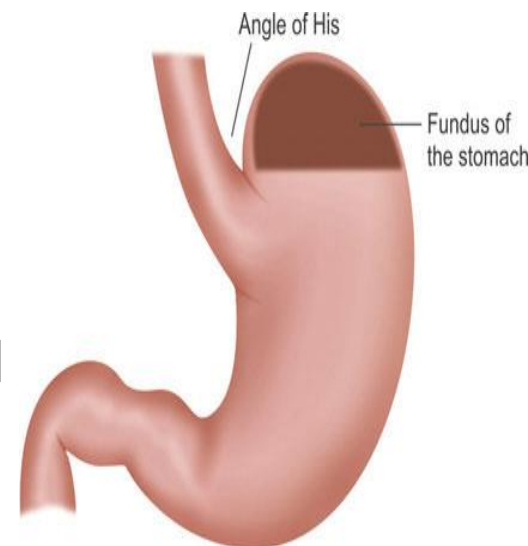


Crural Diaphragm

- The diaphragm provides an important external sphincter as a functional *antireflux barrier*. The esophagus passes from the thorax into the abdomen through an opening in the diaphragm called the esophageal hiatus at T10. The right crus of the diaphragm forms the major muscular component surrounding the esophageal hiatus. In a hiatal hernia, part of the stomach and/or gastroesophageal junction can move through the esophageal hiatus into the thorax.



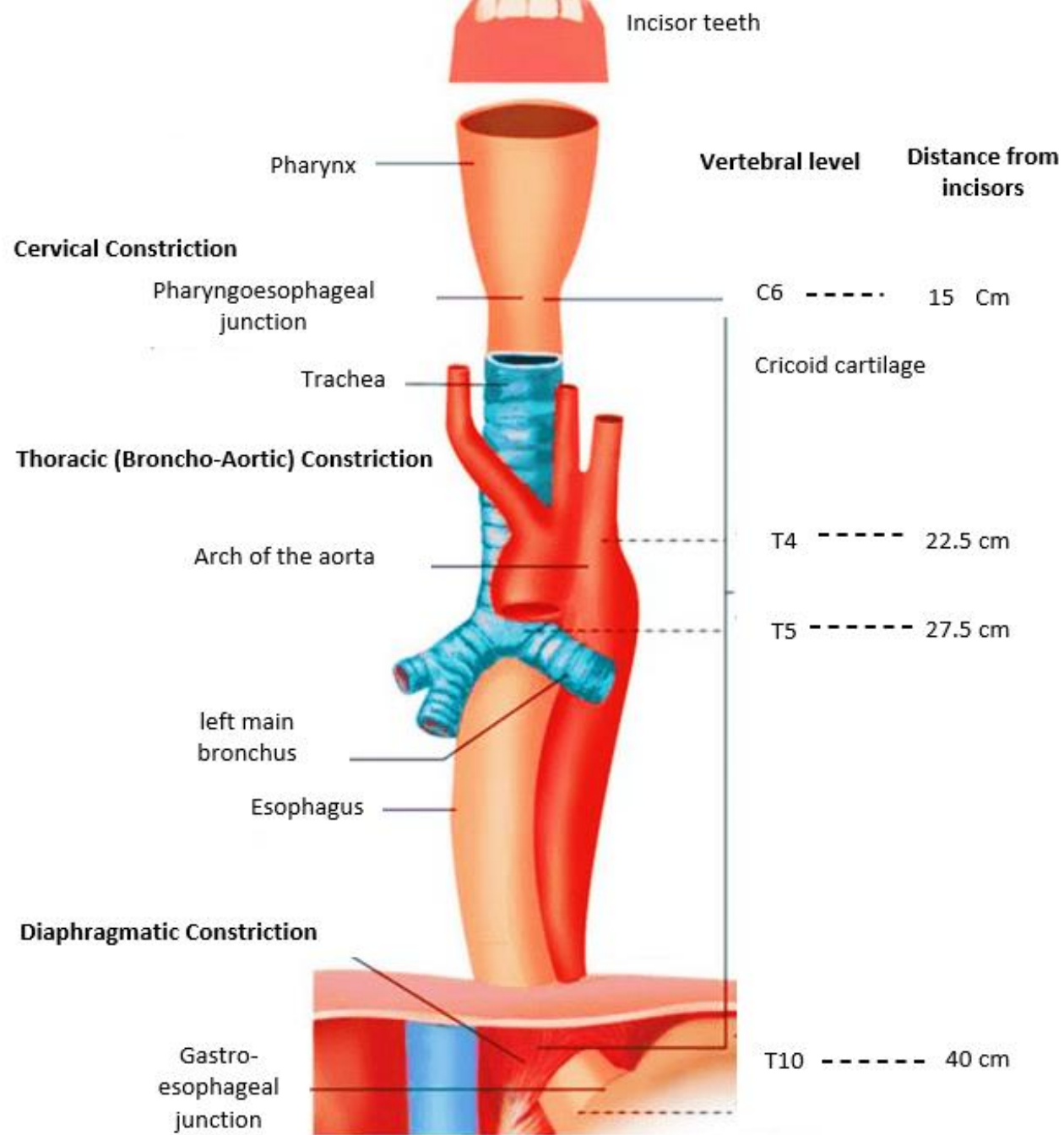
- The angle of His is the acute angle formed between the left side of the abdominal esophagus and the fundus of the stomach at the gastroesophageal junction. At the gastroesophageal junction it acts as a flap-valve mechanism at the gastroesophageal junction.



Esophageal Constriction:

❑ The esophagus has four areas where the lumen is narrower than in other sections

1. **Cervical Constriction:** Located at the pharyngoesophageal junction, 15 cm from the incisor teeth; caused by the cricopharyngeus muscle at the level of C6, clinically referred to as the upper esophageal sphincter.
2. **Thoracic (Broncho-Aortic) Constriction:**
 - a. This is a compound constriction where the esophagus is first crossed by the arch of the aorta at about 22.5 cm from the incisor teeth at the level of T4.
 - b. Secondly, it is crossed by the left main bronchus at approximately 27.5 cm from the incisor teeth at the level of T5.
3. **Diaphragmatic Constriction:** This occurs as the esophagus passes through the esophageal hiatus of the diaphragm, located about 40 cm from the incisor teeth at the level of T11.



mnemonic: *ABCD*



✓ *A- Aortic arch*

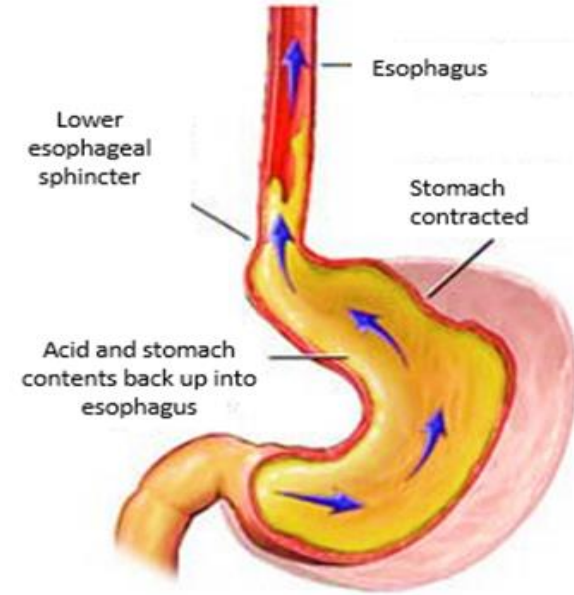
✓ *B- Bronchi (Left)*

✓ *C- Cricopharyngeus*

✓ *D- Diaphragm*

CLINICAL CORRELATION

- **Pyrosis (GERD):** is regurgitated from the stomach into the esophagus due to the dysfunction of the lower esophageal sphincter. This can lead to inflammation and ulcers
- **Achalasia of the Cardia (Esophageal Achalasia):** This condition is characterized by the failure of normal relaxation of the gastroesophageal sphincter, resulting in an obstruction of flow into the stomach.
- **EGD** is used to examine the esophagus, stomach, and duodenum (esophago-gastro-duodenoscopy). This procedure allows visualization of the internal microscopic structures down to the beginning of the third part of the duodenum. A biopsy can be performed, and esophageal varices can be diagnosed. The true length of the esophagus is 25 cm, measured 40 cm from the incisor teeth during the EGD. This information is valuable for diagnosis.





CLINICAL

CASE

SCENARIO

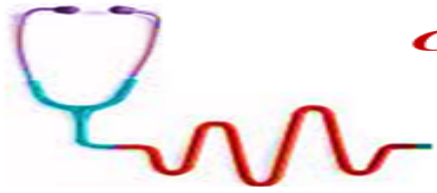


A 60-year-old patient with esophageal carcinoma undergoes evaluation. The cervical part of the esophagus drains lymph primarily into which group of nodes?

- A. Axillary lymph nodes
- B. Deep cervical lymph nodes
- C. Tracheobronchial lymph nodes
- D. Paratracheal lymph nodes
- E. Submandibular lymph nodes

A 52-year-old male with liver cirrhosis develops esophageal varices. The lower esophagus shows dilated veins. The venous drainage of this esophageal segment partly enters the portal circulation. Which vein drains into the portal system from the abdominal part of the esophagus?

- A. Azygos vein
- B. Hemiazygos vein
- C. Left gastric vein
- D. Inferior vena cava
- E. Superior vena cava



CLINICAL

CASE

SCENARIO



A patient undergoing endoscopy shows narrowing of the esophagus at 27.5 cm from the incisors, caused by external compression. Which structure is responsible for this constriction?

- A. Arch of the aorta
- B. Left main bronchus
- C. Right main bronchus
- D. Diaphragmatic hiatus
- E. Inferior pharyngeal constrictor

A 52-year-old man with a history of chronic liver disease presents with vomiting of a large amount of blood. Endoscopy reveals dilated, tortuous veins in the lower part of the esophagus. Which anatomical venous connection is mainly responsible for the formation of esophageal varices?

- A) Superior vena cava and inferior vena cava
- B) Portal vein and azygos venous system
- C) Splenic vein and renal vein
- D) Inferior mesenteric vein and inferior vena cava
- E) Hepatic veins and portal vein

Thank You